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## Embedded Stem Priming Effects in Prefixed and Suffixed Pseudowords

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### ABSTRACT

Previous research has repeatedly revealed evidence for morpho-orthographic priming effects in suffixed words. However, evidence for the morphological chunking of prefixed words is sparse and ambiguous. The goal of the present study was to directly contrast the processing of prefixed and suffixed pseudowords within the same experiment. We carried out a masked primed lexical decision experiment, in which the same target (*AMOUR* [LOVE]) was preceded by a prefixed (*préamour* [prelove]), a nonprefixed (*brosamour* [broslove]), a suffixed (*amoureuse* [lovedom]), and a nonsuffixed (*amourugne* [lovedel]) prime. The results revealed significant priming across all four conditions. Moreover, priming was modulated by individual differences in reading proficiency. High-proficiency readers showed evidence for embedded stem priming effects, independent of whether stems occurred in combination with a real affix or a nonaffix. This finding is of relevance to recent morphological processing theories, suggesting that embedded stems represent salient activation units during the reading of complex pseudowords.

For many years, researchers have explored the mechanisms involved in the visual recognition of morphologically complex words (Taft & Forster, 1975). By now, it is a well-established fact that not only suffixed words (*walker*) but also “pseudosuffixed” words (*corner*) are decomposed into morphemic subunits (*walk-er*; *corn-er*) during visual word recognition (Beyersmann et al., 2015; Feldman, O'Connor, & Moscoso Del Prado Martin, 2009; Rastle, Davis, & New, 2004). Evidence comes primarily from masked morphological priming (for reviews, see Amenta & Crepaldi, 2012; Rastle & Davis, 2008), showing that the recognition of a monomorphemic target word is facilitated by the prior presentation of a related suffixed (*walker-WALK*) or pseudosuffixed prime (*corner-CORN*), relative to an orthographic control (*cashew-CASH*).

Critically, a recent key finding suggests that morpho-orthographic priming effects are complemented by embedded stem priming effects (Beyersmann, Casalis, Ziegler, & Grainger, 2015). Beyersmann and colleagues compared targets (*TRISTE*) preceded by a suffixed word (*tristesse*), a suffixed pseudoword (*tristerie*), a nonsuffixed pseudoword (*tristald*), or an unrelated prime (*direction*). The results revealed priming not only in the two suffixed conditions but also in the nonsuffixed condition, suggesting that embedded stems were activated independently of whether they occurred in combination with a real suffix or a nonsuffix (see also Hasenäcker, Beyersmann, & Schroeder, 2015; Morris, Porter, Grainger, & Holcomb, 2011). Contrary to previous assumptions that morphological decomposition is triggered by the presence of an affix (Taft & Forster, 1975), these data suggest that the presence of an affix is *not* obligatory for the successful activation of the stem's representation. Although Beyersmann et al.'s findings revealed that stems are activated when embedded at word-beginnings (as in suffixed pseudowords), it is not clear whether the same

activation mechanisms apply to stems embedded at word endings (as in prefixed pseudowords). Previous research from monomorphemic word recognition suggest that embedded words are identified independently of position ('hat' in 'hatch'; Bowers, Davis, & Hanley, 2005; see also Crepaldi, Rastle, Davis, & Lupker, 2013, for evidence from pseudocompound processing). However, it is yet to be established if similar principles apply to the processing of prefixed letter strings.

Our goal was to further explore the role of affix and embedded stem activation mechanisms during the processing of prefixed and suffixed letter strings. Although evidence for morpho-orthographic decomposition comes primarily from studies using *suffixed* words, it is not clear whether the same mechanisms apply in the context of *prefixed* words. Pseudoprefixed priming effects in Russian and Dutch suggest that prefixed words, just like suffixed words, are decomposed into morpho-orthographic units (Diependaele, Sandra, & Grainger, 2009; Kazanina, 2011). However, a considerable number of studies have found differences between the processing of suffixed and prefixed words (Beauvillain, 1996; Bergman, Hudson, & Eling, 1988; Beyersmann, Ziegler, & Grainger, 2015; Colé, Beauvillain, Pavard, & Segui, 1986; Colé, Beauvillain, & Segui, 1989; Feldman & Larabee, 2001; Giraudo & Grainger, 2003; Grainger, Colé, & Segui, 1991; Kim, Wang, & Taft, 2015; Meunier & Segui, 2002), and some have failed to find evidence for the morphological decomposition of prefixed words (Colé et al., 1989; Manelis & Tharp, 1977) and prefixed pseudowords (Beyersmann et al., 2015; Kim et al., 2015).

It has been argued that differences between prefixed and suffixed word processing may be due to a left-to-right processing bias in word recognition (Beauvillain, 1996; Colé et al., 1989; Taft, 1979). For instance, an eye-tracking study by Beauvillain (1996) revealed that the processing of prefixed and suffixed words is influenced by the position of the stem. Cumulative stem frequency affected the *second* fixation duration in prefixed words (where the stem follows the affix), but the *first* fixation duration in suffixed words (where the stem precedes the affix). Moreover, prefixes, as opposed to suffixes, have an exclusively semantic function (Bergman et al., 1988; Colé et al., 1989; Giraudo & Grainger, 2003; Libben, 1994). Suffixes can change the syntactic category of the stem, whereas prefixes typically do not. Suffixes can also change the phonological or orthographic form of the stem, whereas prefixes do not. However, the majority of studies examining differences between prefixed and suffixed word processing have used simple lexical decision or overt priming tasks that are open to explicit processing (Beauvillain, 1996; Bergman et al., 1988; Colé et al., 1986; Colé et al., 1989; Feldman & Larabee, 2001; Meunier & Segui, 2002). As a result, they are less able to provide insight into if and how prefixed and suffixed words differ at the early, more automatized stages of visual word recognition. Moreover, previous research has focused on the examination of prefixed *words* (Diependaele et al., 2009; Giraudo & Grainger, 2003; Kazanina, 2011). To our knowledge, no evidence exists to date on the exploration of prefixed *pseudoword* priming effects within Roman alphabetic languages (but see Kim et al., 2015, for recent evidence from Korean). It can therefore not be ruled out that morpho-orthographic decomposition is triggered only when the prefixed string is a real word (for a related hypothesis, see Beyersmann et al., 2015).

Our goal was to directly compare prefixed and suffixed pseudowords with identical stems using masked priming. Pseudowords are particularly useful for the exploration of prelexical effects of morphological segmentation, without lexical interference from the whole word. Moreover, it has been criticised that previously tested pseudosuffixed words vary in the extent to which the suffix conveys regular meaning (Baayen et al., 2011). We avoided this problem by using complex pseudoword primes. A further key benefit is that the same targets could be used across conditions (*préamour* vs. *amouresse*). We can therefore rule out that differences across conditions are due to uncontrolled differences between target words.

If it is true that embedded stems are activated during prefixed pseudoword processing, just like in suffixed pseudoword processing (Beyersmann et al., 2015), we would expect priming independently of whether the target is embedded in an affixed (*préconseil*, where *pré* is a French affix) or a nonaffixed pseudoword (*bréconseil*, where *bré* is not an affix). Moreover, if morpho-orthographic

chunking of prefixes depends on purely structural information, just like the chunking of suffixes (Longtin & Meunier, 2005; Rastle & Davis, 2008), morphological priming should occur with both suffixed and prefixed pseudoword primes. It is important to note that morpho-orthographic chunking effects would be evidenced by significantly greater priming effects in the affixed relative to the nonaffixed conditions.

A second goal was to test if priming is modulated by individual differences in reading proficiency. Most previous research has compared poor with average or good readers to identify deficits that might be responsible for poor readers' failure to develop reading proficiency (Blachman, 1997; Stanovich, 2000). Individual differences among skilled readers have received considerably less attention. Yet there is substantial variability among such samples on a variety of written language proficiency measures (Andrews, 2008; Andrews & Bond, 2009; Andrews & Hersch, 2010; Andrews & Lo, 2012). For instance, Beyersmann et al. (2015) showed that high-proficiency readers are more expert in mapping input letter strings onto orthographic whole-word representations, whereas low-proficiency readers are more reliant on morpho-orthographic segmentation mechanisms (see also Beyersmann, Grainger, Casalis, & Ziegler, 2015, for a comparable pattern in developing readers). We thus included individual average response time as a covariate in the analyses to examine whether individual differences in reading proficiency would modulate priming. Reading skills are inversely correlated with response time in the lexical decision task (i.e., shorter response times with higher reading proficiency: Hargreaves, Pexman, Zdrazilova, & Sargious, 2012; see also Chateau & Jared, 2000, for related evidence on the influence of sublexical skills on reading latencies). We expected that high-proficiency readers would show priming in both affixed *and* nonaffixed conditions, whereas low-proficiency readers would show priming only with affixed, but not nonaffixed pseudowords.

## Method

### Participants

Sixty undergraduate students (1st to 3rd year, *M* age = 19.8) from Aix-Marseille University, all French native speakers, participated for monetary reimbursement (€5/30 min).

### Materials

Forty-eight word targets were selected from Lexique (New, Pallier, Brysbaert, & Ferrand, 2004). Target words were five to seven letters long, were monomorphemic, and ended in a consonant. Each target (*AMOUR*) was preceded by six different primes: a prefixed (*préamour*), nonprefixed (*brosamour*), prefixed control (*brossirgle*), suffixed (*amouresse*), nonsuffixed (*amourugne*), and a suffixed control prime (*broncugne*).

Prefixed and suffixed pseudoword primes were created by combining targets with affixes, such that the whole string was not a word. Nonaffixed pseudoword primes were created by combining the stem with a nonmorphemic ending (*-ugne*, *bré-*, etc.). Unrelated primes were all-letter-different from the target and were created by combining pseudoword stems with nonaffixes (*bronc-ugne*). All pseudowords were orthographically legal and pronounceable. The six prime conditions were matched on prime length and affix/nonaffix length (Appendix A).

Forty-eight pseudoword targets were included for the purpose of the lexical decision task. Pseudowords were created by replacing one to three letters in a baseword (*floir*). Primes preceding pseudoword targets were created based on the same principles as those preceding word targets and comprised the exact same affixes and nonaffixes (*floirire*, *floiroche*, *chippire*, *refloir*, *gefloir*, *rechipp*). Pseudoword and word targets, as well as their corresponding primes, were matched on length. To avoid that participants saw any target twice, we created six counterbalanced lists.

## Procedure

Stimuli were presented in the centre of a CRT computer screen, using DMDX (Forster & Forster, 2003). Each trial consisted of a 500-ms forward mask of hash keys, then a 50-ms prime in lowercase, then the uppercase target. The target remained present until the response or until 3 s had elapsed. Participants were instructed to respond as quickly and accurately as possible.

## Results

Lexical decisions to word targets were analysed as follows. Incorrect responses were removed from the reaction time (RT) analysis (2.3% of all data). RTs were 1/RT transformed. Transformed RTs smaller than  $Q1 - (2.5 \times IQR)$  or larger than  $Q3 + (2.5 \times IQR)$ , by either participants or items (1.1% of the data), were set to the identified upper and lower boundaries ( $Q1$  = first quartile,  $Q3$  = third quartile, and  $IQR = Q3 - Q1$  = interquartile range). RTs and error rates were analysed for each subject (Table 1).

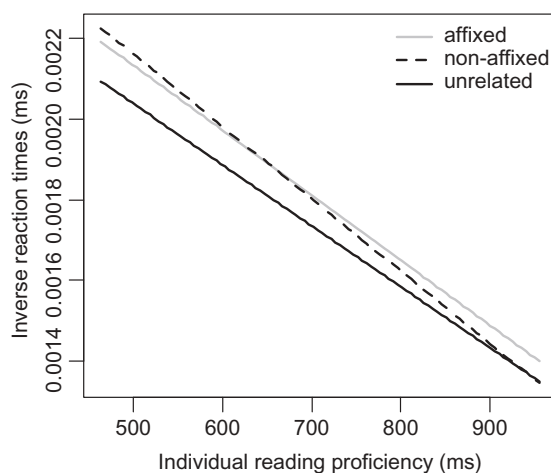
We used linear mixed-effect model modelling to perform the main analyses (Baayen, 2008; Baayen, Davidson, & Bates, 2008). Fixed and random effects were included only if they significantly improved the model's fit in a backward stepwise model selection procedure. Models were selected using chi-square log-likelihood ratio tests with regular maximum likelihood parameter estimation. Trial order was included to control for longitudinal task effects such as fatigue or habituation. Chunk-frequency was included as a continuous factor to control for frequency differences between strings that formed affixes compared to strings that formed nonaffixes. Stems were classified depending on their position (ending: as in prefixed and nonprefixed items; beginning: as in suffixed and nonsuffixed items). Individual reading proficiency was calculated based on the mean RT for each participant across word and nonword trials. A linear mixed-effects model as implemented in the lme4 package (Bates, Maechler, Bolker, & Walker, 2014) in the statistical software R (Version 3.0.3; RDevelopmentCoreTeam, 2008) was created with five fixed effects factors (prime type: affixed, nonaffixed, unrelated; position: beginning, ending; chunk-frequency; individual reading proficiency; trial order), their interactions, and two random effects factors (random intercepts for subjects and items). The model was refitted after excluding data points with standardised residuals larger than 2.5 in absolute value (see Baayen, 2008). *T* values, *z* values, *p* values, and standard errors were determined using the package lmerTest (Kuznetsova, Brockhoff, & Christensen, 2014).

The RT analyses revealed significant priming in the affixed and nonaffixed conditions ( $t = 2.66$ ,  $p = .008$  and  $t = 3.45$ ,  $p < .001$ ), but no difference between the affixed and nonaffixed conditions ( $t = 0.97$ ,  $p = .429$ ). Critically, priming in the nonaffixed condition was modulated by individual reading proficiency ( $t = 2.45$ ,  $p < .014$ ), suggesting that priming increased with increasing reading

**Table 1.** RTs (in ms) and ER%, averaged across items for each participant.

| Condition                      | RTs      | ER%       | Example                   |
|--------------------------------|----------|-----------|---------------------------|
| Stem embedded at the ending    |          |           |                           |
| Affixed                        | 542 (76) | 2.5 (6.4) | <i>préamour-AMOUR</i>     |
| Nonaffixed                     | 543 (76) | 1.9 (4.5) | <i>brosamour-AMOUR</i>    |
| Unrelated                      | 562 (79) | 3.1 (7.5) | <i>brossirgle-AMOUR</i>   |
| Stem embedded at the beginning |          |           |                           |
| Affixed                        | 534 (71) | 1.5 (4.7) | <i>amouresse-AMOUR</i>    |
| Nonaffixed                     | 531 (28) | 2.1 (4.7) | <i>amourugne-AMOUR</i>    |
| Unrelated                      | 559 (72) | 2.9 (2.6) | <i>broncugne-AMOUR</i>    |
| Average across all trials      |          |           |                           |
| Affixed                        | 538 (71) | 2.0 (4.1) | <i>see examples above</i> |
| Nonaffixed                     | 537 (71) | 2.0 (3.9) | <i>see examples above</i> |
| Unrelated                      | 560 (70) | 3.0 (5.2) | <i>see examples above</i> |

Note. Standard deviations are in parentheses. Significant priming effects were observed in the affixed and nonaffixed conditions (relative to the unrelated control condition), independent of whether the target word was embedded at the beginning or the end of the prime. RT = reaction time; ER% = error rate.



**Figure 1.** Inverse reaction times to targets preceded by affixed, nonaffixed, and unrelated pseudoword primes, as a function of individual reading proficiency. *Note.* Significant priming in the affixed condition was observed across all participants (relative to the unrelated control), whereas priming in the nonaffixed condition was modulated by individual reading speed, with fast readers showing more priming than slow readers.

proficiency (Figure 1). There was also a significant main effect of individual reading proficiency ( $t > 10$ ,  $p < .001$ ). Moreover, we observed a significant position effect in the affixed and nonaffixed conditions ( $t = 2.16$ ,  $p = .031$  and  $t = 2.56$ ,  $p = .011$ ) but not in the unrelated control condition ( $t = 0.38$ ,  $p = .703$ ), suggesting that participants responded faster when targets were embedded at the beginning than at the ending of the prime. However, the interaction between prime type and position was not significant (see Appendix B for a detailed output of the linear mixed-effects analyses), thus making it difficult to univocally attribute the position effect to the embedded stems. No other effects were significant.

Error analyses followed the same logic as the RT analyses. We applied a binomial variance assumption to the trial-level binary data using the function *glmer* as part of the R-package *lme4*. There was a significant effect of individual reading proficiency ( $z = 3.36$ ,  $p < .001$ ). No other effects were significant.

## Discussion

The study's goal was to explore differences in the processing of prefixed and suffixed pseudowords during the early stages of visual word recognition. Most important, to provide a more tightly controlled comparison of embedded stem priming effects in prefixed and suffixed pseudowords, we selected targets that were *identical* in both prefixed and suffixed conditions. The results revealed significant and equal priming effects in the affixed and nonaffixed conditions, relative to the unrelated control. Priming effects were not modulated by position, suggesting that the identification of stem morphemes is position-independent (Crepaldi et al., 2013). Our results are consistent with a recent key finding, suggesting that embedded stems are activated independently of whether they occur in combination with a real suffix or a nonsuffix (Beyersmann et al., 2015; Morris et al., 2011). Our data extend this important finding to the context of prefixed letter strings, suggesting that stems are activated not only when embedded at word beginnings but also when embedded at word endings. These findings are of great importance to theories of morphological processing. For many years, it has been believed that morphological decomposition is triggered by the presence of an affix (Taft & Forster, 1975). Only after the affix has been identified and stripped off are the letters of the stem mapped onto a higher level lexical representation. Our data suggest that affix-stripping is clearly insufficient to account for the present findings, as our data unambiguously demonstrate that



embedded stems are activated during the early stages of visual word recognition, even in the absence of an affix.

Our second key finding is that priming in the nonaffixed condition was modulated by individual differences in reading proficiency, suggesting that nonaffixed priming increased with increasing reading proficiency. Embedded stem priming effects (i.e., comparable magnitudes of priming in both affixed and nonaffixed conditions) were particularly pronounced in high-proficiency participants (Figure 1), suggesting that these individuals are more proficient in mapping the letters of embedded stems onto orthographic whole-word representations. In contrast, low-proficiency participants showed robust priming only in the affixed condition, suggesting that these individuals are more reliant on segmentation mechanisms that help to parse the visual input into morphemic substrings. Hence, the different pattern of priming seen in high- versus low-proficiency readers speak in favour of theories suggesting that both whole words and morphemes can be activated in parallel during the initial stages of visual word recognition (Baayen & Schreuder, 1999; Beyersmann, Coltheart, & Castles, 2012; Grainger & Ziegler, 2011; Marelli, Amenta, Morone, & Crepaldi, 2013).

Our findings are consistent with Beyersmann et al. (2015), who reported that participants with an overall higher proficiency in spelling and vocabulary showed more priming in the nonaffixed pseudoword condition than participants with comparatively lower proficiency scores. Interestingly, a *dissociation* between spelling and vocabulary was reported by Andrews and Lo (2013): A greater influence of semantic transparency in processing complex *real* words was observed in participants who had comparatively high vocabulary scores compared with their spelling ability. Participants with higher spelling ability than vocabulary showed comparable magnitudes of true and pseudomorphological priming, suggesting that spelling proficiency assists during efficient morphological chunking. However, it is possible that the automatic segmentation of morphologically structured pseudowords (like the ones we used here) requires different, or perhaps lesser, reading skills than the segmentation of complex real words (like the ones used by Andrews & Lo, 2013), because the activation of the stem is not inhibited by the lexical representation of the whole word. Key questions thus remain concerning the dissociation between spelling and vocabulary, but also regarding the impact of differences in word and pseudoword reading proficiency during morphological processing.

Interestingly, our findings converge with recent developmental data from Beyersmann et al. (2015), who reported increased embedded stem priming effects in high- relative to low-proficiency children. In contrast to adults, however, low-proficiency children showed no evidence for morpho-orthographic priming, which is consistent with other previous failures to observe morpho-orthographic processing in developing readers (Beyersmann, Castles, & Coltheart, 2012; Hasenäcker et al., 2015; but see Quémart, Casalis, & Colé, 2011). This result suggests that basic embedded stem activation mechanisms are acquired relatively early during reading development and are particularly pronounced in individuals with higher levels of reading proficiency, whereas morpho-orthographic processing appears to be a quite late occurring milestone during reading acquisition. The later acquired, more abstract morpho-orthographic representations are presumably added to the reading system, such that whole-word representations remain available in a parallel processing route. Future research is clearly necessary to further examine the interplay between embedded stem and morpho-orthographic activation mechanisms during reading development.

Given the evidence for embedded stem priming effects in the present study, why then did earlier morphological priming studies (for a review, see Rastle & Davis, 2008) fail to provide evidence for embedded stem priming effects within word primes (*cashew-CASH*)? As already discussed in Beyersmann et al. (2015), one explanation for the absence of *cashew-CASH* priming is that previous morpho-orthographic priming experiments did not assess individual levels of language proficiency, such that the *cashew-CASH* effect may have been washed out by the presence of low-proficiency participants. However, a more likely explanation is that competition arises between the target (*cash*) and the prime, which itself is a word (*cashew*), thus preventing stem-target priming. Lexical competition would also occur between pseudo-suffixed primes and their targets (between *corn* and *corner*), but here the activation of the target benefits from the form-based decomposition of the

prime into morpho-orthographic subunits, which explains why stem-target priming does typically arise in this condition (Rastle & Davis, 2008).

Our French results (see also Beyersmann et al., 2015) are consistent with evidence from English (Morris et al., 2011) and German (Hasenäcker et al., 2015), suggesting that in several Indo-European languages, target recognition is facilitated by the prior presentation of a target-embedded prime. However, studies examining Semitic languages (e.g., Hebrew and Arabic) with divergent internal word structures (e.g., stems and affixes are intertwined rather than concatenated linearly) suggest that although morphological priming effects are strong, form-related priming effects are less likely to occur than in other Indo-European languages (Frost, Kugler, Deutsch, & Forster, 2005; Velan & Frost, 2011). It is therefore possible that the presently observed embedded stem priming effects do not generalise to languages such as Hebrew or Arabic.

In conclusion, our data provide unequivocal evidence for the activation of embedded stems at very early prelexical processing stages within the reading system. The observed priming effects reflect the rapid mapping of embedded letter sequences onto whole-word representations in the orthographic lexicon, suggesting that affix-stripping (Taft & Forster, 1975), which for many years has been believed to be the key mechanism underlying morpho-orthographic processing, is not sufficient to account for our present data. Questions remain regarding whether embedded stem activation mechanisms and affix-chunking mechanisms concurrently boost morphological segmentation during the early stages of visual word recognition, or if both mechanisms have competitive functions. A logical sequel of the present study would be the exploration of high-temporal resolution recordings of event-related brain potentials to investigate the time course of stem and affix processing in individuals with varying levels of reading proficiency.

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## Appendix A

**Table A1.** Pseudoword primes and word targets.

| Suffixed    | Nonsuffixed | Unrelated   | Prefixed     | Nonprefixed | Unrelated    | Target  |
|-------------|-------------|-------------|--------------|-------------|--------------|---------|
| conseilleux | conseilluve | peuprerore  | préconseil   | bréconseil  | brépeupre    | conseil |
| hôpitalerie | hôpitalome  | blisquerom  | prohôpital   | drohôpital  | droblisque   | hôpital |
| sommeilloir | sommeillome | vaureclache | intersommeil | fesommeil   | fevaucle     | sommeil |
| travaillade | travaillape | speltterape | intertravail | brétravail  | bréspeltre   | travail |
| chagrinade  | chagrinoche | grobilloche | néochagrin   | gréchagrin  | grégrobille  | chagrin |
| magasinet   | magasinalve | stolchale   | antimagasin  | cismagasin  | coistolche   | magasin |
| compléteux  | completme   | praudrerime | précomplet   | brécomplet  | brépraudre   | complet |
| contenterie | contentarbe | misquerel   | intercontent | tiscontent  | tismisque    | content |
| délicature  | délicatuve  | stumfluve   | antidélicat  | drodélicat  | drostufle    | délicat |
| discretesse | discretiche | blinchiche  | prodiscret   | gediscret   | geblinche    | discret |
| inquiéteux  | inquiétome  | pemillore   | reinquiet    | brosinquiet | brospemille  | inquiet |
| méchantien  | méchantore  | faucrerore  | proméchant   | feméchant   | fefaucre     | méchant |
| violentoir  | violentuve  | broelluve   | revolent     | bréviolent  | brébroelle   | violent |
| hasarditude | hasardore   | chobirache  | préhasard    | gréhasard   | gréchobire   | hasard  |
| secondeute  | secondeuve  | floissuve   | néosecond    | visisecond  | visifloisse  | second  |
| animaleur   | animalugne  | saicrugne   | préanimal    | brosanimal  | brosaire     | animal  |
| chevaleur   | chevalarbe  | jernarbe    | néocheval    | cis cheval  | cisjendre    | cheval  |
| camionnable | camionnope  | traustope   | anticamion   | brécamion   | brétrauste   | camion  |
| cartonnier  | cartonnime  | vuillimime  | anticarton   | brécarton   | brévruime    | carton  |
| cheminade   | cheminome   | glaistome   | interchemin  | drochemin   | droglaiste   | chemin  |
| cousinesse  | cousinond   | stornecond  | néocousin    | gecousin    | gestorne     | cousin  |
| dessinoir   | dessinalve  | crimbrale   | prédessin    | fedessin    | fecrimbre    | dessin  |
| jardinable  | jardinide   | grufride    | antijardin   | bréjardin   | bréfrufré    | jardin  |
| voisinette  | voisinolve  | ploirolve   | prévoisin    | tisvoisin   | tisploindre  | voisin  |
| obscurresse | obscurome   | pluntome    | préobscur    | brosobscur  | brosplunte   | obscur  |
| argentine   | argentele   | claffrave   | désargent    | brosoargent | broscloffre  | argent  |
| combatible  | combatoche  | frauroche   | intercombat  | visicombat  | visifraure   | combat  |
| désertoir   | désertime   | gourplime   | prédésert    | tisdésert   | tisgourple   | désert  |
| enfantif    | enfantore   | vondore     | néoenfant    | brosoenfant | brosoberque  | enfant  |
| espritude   | espritide   | trinpiche   | néoesprit    | brosoesprit | brostrinpe   | esprit  |
| secrétesse  | secretoche  | neosecret   | néosecret    | grésecret   | gréteuple    | secret  |
| froidible   | froidoche   | baugoche    | refroid      | drofroid    | drobaug      | froid   |
| blondesse   | blondime    | teuplime    | interblond   | feblond     | feteuple     | blond   |
| chauderie   | chaudoma    | plimpelord  | antichaud    | bréchaud    | bréplimpte   | chaud   |
| lourdesse   | lourdacre   | deustacre   | antilourd    | grélourd    | grédeuste    | lourd   |
| sourdette   | sourdiche   | cruntiche   | resourd      | visisourd   | visicrunt    | sourd   |
| fusilleur   | fusillope   | douterope   | antifusil    | tisfusil    | tisdout      | fusil   |
| avionnette  | avionnuve   | prenchuve   | proavion     | brosoavion  | brosoprenche | avion   |
| matineux    | matinond    | diurtond    | prématin     | fematin     | fediurt      | matin   |
| amouresse   | amourugne   | broncugne   | préamour     | brosoamour  | brosoirgle   | amour   |
| hiveret     | hiverel     | freupupe    | prohiver     | drohiver    | drofreupe    | hiver   |
| claireur    | clairuto    | rengluto    | interclair   | geclair     | gerengle     | clair   |
| épaisier    | épaissel    | bodefoche   | désépais     | brosoépais  | brosobodefe  | épais   |
| arrétier    | arrétore    | ploroire    | interarrêt   | brosoarrêt  | brosoploi    | arrêt   |
| bruitoir    | bruitugne   | tonurugne   | prébruit     | cisbruit    | cistonur     | bruit   |
| forétain    | forétupe    | brumerupe   | interforêt   | droforêt    | drobrumer    | forêt   |
| saintesse   | saintiche   | frempiche   | présaint     | grésaint    | grécrumpe    | saint   |
| saluteur    | salutel     | clonache    | resalut      | drosalut    | droclon      | salut   |

## Appendix B

**Table B1.** Output of the linear mixed-effects analyses conducted on the reaction time data, exploring the factor of primetype [affixed, nonaffixed, unrelated], position [beginning, ending] and individual reading proficiency [read prof].

| Factor                                  | Estimate  | SE       | df       | t Value | Pr(> t ) |
|-----------------------------------------|-----------|----------|----------|---------|----------|
| <i>primetype reference: unrelated</i>   |           |          |          |         |          |
| (Intercept)                             | 2.82E-03  | 9.63E-05 | 8.36E+01 | 29.26   | < .001   |
| primetype[nonaffixed]                   | 2.24E-04  | 6.48E-05 | 1.82E+03 | 3.45    | < .001   |
| primetype[affixed]                      | 1.73E-04  | 6.49E-05 | 1.82E+03 | 2.66    | .008     |
| read prof                               | -1.54E-06 | 1.48E-07 | 8.03E+01 | -10.43  | < .001   |
| position[ending]                        | 6.20E-06  | 1.63E-05 | 1.82E+03 | 0.38    | .703     |
| primetype[nonaffixed: read prof]        | -2.42E-07 | 9.87E-08 | 1.82E+03 | -2.45   | .014     |
| primetype[affixed: read prof]           | -1.52E-07 | 9.87E-08 | 1.82E+03 | -1.54   | .125     |
| primetype[nonaffixed]: position[ending] | 3.53E-05  | 2.30E-05 | 1.82E+03 | 1.54    | .125     |
| primetype[affixed]: position[ending]    | 2.89E-05  | 2.30E-05 | 1.82E+03 | 1.26    | .210     |
| <i>primetype reference: affixed</i>     |           |          |          |         |          |
| (Intercept)                             | 2.99E-03  | 9.62E-05 | 8.33E+01 | 31.08   | < .001   |
| primetype[unrelated]                    | -1.73E-04 | 6.49E-05 | 1.82E+03 | -2.66   | .008     |
| primetype[nonaffixed]                   | 5.12E-05  | 6.47E-05 | 1.82E+03 | 0.79    | .429     |
| read prof                               | -1.69E-06 | 1.47E-07 | 7.98E+01 | -11.48  | < .001   |
| position[ending]                        | 3.51E-05  | 1.63E-05 | 1.82E+03 | 2.16    | .031     |
| primetype[unrelated: read prof]         | 1.52E-07  | 9.87E-08 | 1.82E+03 | 1.54    | .125     |
| primetype[nonaffixed: read prof]        | -9.02E-08 | 9.85E-08 | 1.82E+03 | -0.92   | .360     |
| primetype[unrelated]: position[ending]  | -2.89E-05 | 2.30E-05 | 1.82E+03 | -1.26   | .210     |
| primetype[nonaffixed]: position[ending] | 6.42E-06  | 2.30E-05 | 1.82E+03 | 0.28    | .780     |
| <i>primetype reference: nonaffixed</i>  |           |          |          |         |          |
| (Intercept)                             | 3.04E-03  | 9.62E-05 | 8.31E+01 | 31.63   | < .001   |
| primetype[affixed]                      | -5.12E-05 | 6.47E-05 | 1.82E+03 | -0.79   | .429     |
| primetype[unrelated]                    | -2.24E-04 | 6.48E-05 | 1.82E+03 | -3.45   | < .001   |
| read prof                               | -1.78E-06 | 1.47E-07 | 7.99E+01 | -12.09  | < .001   |
| position[ending]                        | 4.15E-05  | 1.62E-05 | 1.82E+03 | 2.56    | .011     |
| primetype[affixed: read prof]           | 9.02E-08  | 9.85E-08 | 1.82E+03 | 0.92    | .360     |
| primetype[unrelated: read prof]         | 2.42E-07  | 9.87E-08 | 1.82E+03 | 2.45    | .014     |
| primetype[affixed]: position[ending]    | -6.42E-06 | 2.30E-05 | 1.82E+03 | -0.28   | .780     |
| primetype[unrelated]: position[ending]  | -3.53E-05 | 2.30E-05 | 1.82E+03 | -1.54   | .125     |